

1. More On Planck's Blackbody Formula

a)

To maximize $\epsilon_{Planck}(\lambda)$, we have the condition

$$\frac{d\epsilon_{Planck}}{d\lambda} = 0$$

Set $\frac{d\epsilon}{d\lambda} = 0$. Let $u = \frac{hc}{\lambda k_B T}$, so $\lambda = \frac{hc}{u k_B T}$

The condition $\frac{d\epsilon}{d\lambda} = 0$ becomes (after algebra):

$$\frac{d}{d\lambda} [\lambda^{-5} (e^{hc/\lambda k_B T} - 1)^{-1}] = 0$$

Computing the derivative and setting it to zero:

$$-5\lambda^{-6}(e^u - 1)^{-1} + \lambda^{-5} \cdot \frac{e^u}{(e^u - 1)^2} \cdot \frac{hc}{\lambda^2 k_B T} = 0$$

Multiply through by $\lambda^6(e^u - 1)$:

$$-5(e^u - 1) + u e^u = 0$$

This gives the transcendental equation:

$$(5 - u)e^u = 5$$

This must be solved numerically. Using a calculator, we can iterate and get $u \approx 4.96511$

Then $\lambda_{peak} T = \frac{hc}{u k_B}$:

$$\lambda_{peak} T = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{(4.96511)(1.381 \times 10^{-23})} = \frac{1.988 \times 10^{-25}}{6.857 \times 10^{-23}} \approx 2.898$$

b)

The total intensity must be the same whether integrated over λ or ω :

$$I = \int_0^\infty d\lambda \epsilon(\lambda) = \int_0^\infty d\omega \epsilon(\omega)$$

To solve this, we use the substitution $\omega = \frac{2\pi c}{\lambda}$, so $\lambda = \frac{2\pi c}{\omega}$ and $|\frac{d\lambda}{d\omega}| = \frac{2\pi c}{\omega^2}$

Substituting into $\epsilon(\lambda)$:

$$\epsilon(\omega) d\omega = \epsilon(\lambda) \left| \frac{d\lambda}{d\omega} \right| d\omega = \frac{2\pi h c^2}{\left(\frac{2\pi c}{\omega}\right)^5} \cdot \frac{1}{e^{\hbar\omega/k_B T} - 1} \cdot \frac{2\pi c}{\omega^2} d\omega$$

The λ^5 term becomes $\frac{(2\pi c)^5}{\omega^5}$. Substituting $h = 2\pi\hbar$:

$$\epsilon(\omega) = \frac{2\pi \cdot 2\pi\hbar \cdot c^2 \cdot \omega^5}{(2\pi c)^5} \cdot \frac{1}{e^{\hbar\omega/k_B T} - 1} = \frac{\hbar\omega^3}{(2\pi c)^2} \cdot \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

$$\epsilon_{Planck}(\omega) = \frac{\hbar\omega^3}{(2\pi c)^2} \cdot \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

c)

To solve the integral we use the substitution,

$$u = \frac{\hbar\omega}{k_B T}$$

rewriting the integral,

$$I = \int_0^\infty \frac{\hbar}{(2\pi c)^2} \left(\frac{u k_B T}{\hbar} \right)^3 \frac{1}{e^u - 1} \frac{k_B T}{\hbar} du = \int_0^\infty \frac{(k_B T)^4}{(2\pi c)^2 \hbar^3} \frac{u^3}{e^u - 1} = \frac{(k_B T)^4}{(2\pi c)^2 \hbar^3} \frac{\pi^4}{15} = \sigma T^4$$

$$\sigma = \frac{(k_B \pi)^4}{15(2\pi c)^2 \hbar^3} = \frac{2\pi^5 k_B^4}{15 h^3 c^2}$$

2. The Uncertainty Principle with Gaussian Wave Packets

a)

Since $|\psi|^2 = (2\pi\sigma^2)^{-1/2} e^{-x^2/2\sigma^2}$ is a Gaussian centered at zero:

$$\langle x \rangle = 0$$

$$\langle x^2 \rangle = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^2 e^{-x^2/2\sigma^2} dx = \sigma^2$$

using the standard Gaussian moment $\int x^2 e^{-x^2/2\sigma^2} dx = \sqrt{2\pi}\sigma^3$

$$\Delta x = \sigma$$

b)

$$\tilde{\psi}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x, 0) e^{-ikx} dx$$

Substituting $\psi(x, 0)$,

$$\tilde{\psi}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/4\sigma^2} e^{ik_0 x} e^{-ikx} dx$$

we can take out the normalization factor outside the integral

$$\tilde{\psi}(k) = \frac{1}{\sqrt{2\pi}} \frac{1}{(2\pi\sigma^2)^{1/4}} \int_{-\infty}^{\infty} e^{-x^2/4\sigma^2 + i(k_0 - k)x} dx$$

Now we complete the square in the exponent:

$$-\frac{x^2}{4\sigma^2} - i(k - k_0)x = -\frac{1}{4\sigma^2} [x + 2i\sigma^2(k - k_0)]^2 - \sigma^2(k - k_0)^2.$$

The Gaussian integral gives $\int e^{-(x+const)^2/4\sigma^2} dx = 2\sigma\sqrt{\pi}$:

$$\tilde{\psi}(k) = \frac{2\sigma\sqrt{\pi}}{(2\pi\sigma^2)^{1/4}\sqrt{2\pi}} e^{-\sigma^2(k-k_0)^2} = \left(\frac{2\sigma^2}{\pi}\right)^{1/4} e^{-\sigma^2(k-k_0)^2}$$

$$\tilde{\psi}(k) = \left(\frac{2\sigma^2}{\pi}\right)^{1/4} \exp[-\sigma^2(k-k_0)^2]$$

This is itself a Gaussian in k -space centered on k_0 , with standard deviation $\frac{1}{2\sigma}$.

c)

$|\tilde{\psi}(k)|^2 = \left(\frac{2\sigma^2}{\pi}\right)^{1/2} e^{-2\sigma^2(k-k_0)^2}$ is a Gaussian in k centered at k_0 with variance $\frac{1}{4\sigma^2}$

$$\langle k \rangle = k_0, \langle k^2 \rangle - \langle k \rangle^2 = \frac{1}{4\sigma^2}$$

$$\Delta k = \frac{1}{2\sigma}, \Delta p = \hbar \Delta k = \frac{\hbar}{2\sigma}$$

d)

$$\Delta x \cdot \Delta p = \sigma \cdot \frac{\hbar}{2\sigma} = \frac{\hbar}{2}$$

$$\Delta x \Delta p = \frac{\hbar}{2} \geq \frac{\hbar}{2}$$

The Gaussian saturates the Heisenberg bound, it is the minimum uncertainty state. No other normalised wave packet achieves a smaller product of $\Delta x \Delta p$.